YUWEI ZHANG

UC San Diego ◇ 9500 Gilman Drive, La Jolla, CA 92093-0021
zhangyuwei.work@gmail.com ◇ https://zhang-yu-wei.github.io/

EDUCATION

University of California, San Diego, San Diego, CA, U.S. *Sept, 2023 – 2027*
Degree: Ph.D. in Data Science
Advisor: Prof. Jingbo Shang

University of California, San Diego, San Diego, CA, U.S. *Sept, 2021 – Jun, 2023*
Degree: M.S. in Electrical Engineering
Track: Machine Learning & Data Science

Nankai University, Tianjin, P.R. China *Aug, 2016 – Jun, 2020*
Degree: B.S. in Physics

INDUSTRY EXPERIENCE

Google Deepmind, CA, U.S. *summer 2026*
Job Title: Student Researcher
Project: Agent Post-Training
Results: raised agent pass@1 by about 4 points while cutting tokens per episode by about 60%

Amazon Rufus (Foundation Models Team), CA, U.S. *summer 2025 - spring 2026*
Job Title: Applied Scientist Intern
Project: Agent Post-Training and Context Management
Papers: CoMem, RESD, PATR, SWAP

AMD Research, CA, U.S. *spring 2025*
Job Title: Research Scientist Intern
Project: Reinforcement Learning for VLMs

Tencent America, WA, U.S. *summer 2024*
Job Title: Research Scientist Intern
Project: Long-Term Memory and Knowledge Injection
Papers: LongMemEval, WikiDYK

Amazon AWS, WA, U.S. *summer 2023*
Job Title: Applied Scientist Intern
Project: Representation Learning
Papers: Intent Semantic Toolkit

SELECTED PUBLICATIONS

* Equal contribution.

Reinforcement Learning and Agent Post-Training

- **RESD.** Yuwei Zhang, Sha Li, Changlong Yu, Qin Lu, Shuowei Jin, Chengyu Dong, Haoran Liu, Ilgee Hong, Xintong Li, Zhenyu Shi, Bing Yin, Jingbo Shang. Learning with Rare Success but Rich Feedback via Reflection-Enhanced Self-Distillation. *2026.* [paper] [blog post] [code] **A new paradigm pairing recursive self-improving context with self-distillation: +25.5 points on Manufactoria.**

- **HERO.** Yuwei Zhang*, Haoran Liu*, Xiyao Li, Bohan Lyu, Jingbo Shang. HERO: Hindsight-Enhanced Reflection from Environment Observations for Agentic Self-Distillation. *2026.* [paper] [code] **Upgrades on-policy self-distillation for agents: 28% fewer turns than GRPO on τ-Bench.**
- **PATR.** Xintong Li, Sha Li, Yuwei Zhang, Changlong Yu, Rongmei Lin, Hongye Jin, Shuyi Guan, Xin Liu, Linwei Li, Qingyu Yin, Jingbo Shang. Process Reward Informed Tree Rollout for Effective Multi-Turn RL. *2026.* [paper] **Upgrades RL rollouts with tree search over promising states: +5.0 on SWE-Bench.**
- **SWAP.** Xintong Li, Sha Li, Rongmei Lin, Hongye Jin, Linwei Li, Hejie Cui, Sarah Zhang, Chia-Yuan Chang, Kewei Cheng, Yuwei Zhang, Qingyu Yin, Jingbo Shang, Bing Yin. Stepwise Penalization for Length-Efficient Chain-of-Thought Reasoning. *NeurIPS 2026.* [paper] **Upgrades RL rollouts with step-level length penalties: 64.3% shorter reasoning at +5.7% accuracy.**
- **MaPPO.** Guangchen Lan, Sipeng Zhang, Tianle Wang, Yuwei Zhang, Daoan Zhang, Xinpeng Wei, Xiaoman Pan, Hongming Zhang, Dong-Jun Han, Christopher G. Brinton. MAPPO: Maximum a Posteriori Preference Optimization with Prior Knowledge. *NeurIPS 2026.* [paper] **Upgrades DPO: up to +12.3 on AlpacaEval 2.0 and +13.7 on Arena-Hard.**
- **ChipMATE.** Zhongkai Yu, Yichen Lin, Chenyang Zhou, Yuwei Zhang, Kun Zhou, Junxia Cui, Haotian Ye, Zhengding Hu, Zaifeng Pan, Ruiyi Wang, Yujie Zhao, Hejia Zhang, Jingbo Shang, Jishen Zhao, Yufei Ding. ChipMATE: Multi-Agent Training via Reinforcement Learning for Enhanced RTL Generation. *2026.* [paper] [code] **First self-trained multi-agent framework for RTL generation, the code that specifies chip hardware.**

Memory, Long Context, and Knowledge

- **CoMem.** Yuwei Zhang, Chengyu Dong, Shuowei Jin, Changlong Yu, Hejie Cui, Hongye Jin, Xinyang Zhang, Hamed Bonab, Colin Lockard, Jianshu Chen, Zhenyu Shi, Jingbo Shang, Xian Li, Bing Yin. CoMem: Context Management with A Decoupled Long-Context Model. *ICML 2026.* [paper] [code] **A new compaction framework: up to 2.1× faster, and above full-context accuracy on DeepSWE.**
- **WikiDYK.** Yuwei Zhang, Wenhao Yu, Shangbin Feng, Yifan Zhu, Letian Peng, Jayanth Srinivasa, Gaowen Liu, Jingbo Shang. Bidirectional LMs are Better Knowledge Memorizers? A Benchmark for Real-World Knowledge Injection. *ACL 2026.* [paper] [code] **Bidirectional LMs beat causal LMs at memorization.**
- **LongMemEval.** Di Wu, Hongwei Wang, Wenhao Yu, Yuwei Zhang, Kai-Wei Chang, Dong Yu. LongMemEval: Benchmarking Chat Assistants on Long-Term Interactive Memory. *ICLR 2025.* [paper] **Adopted by Mem0, Zep and ByteRover as a standard evaluation; 1.1k GitHub stars.**
- **Speculative RAG.** Zilong Wang, Zifeng Wang, Long Le, Huaixiu Steven Zheng, Swaroop Mishra, Vincent Perot, Yuwei Zhang, Anush Mattapalli, Ankur Taly, Jingbo Shang, Chen-Yu Lee, Tomas Pfister. Speculative RAG: Enhancing Retrieval Augmented Generation through Drafting. *ICLR 2025.* [paper] **A multi-agent drafter–verifier framework: +12.97% accuracy and 50.83% lower latency.**

TEACHING

TA Fall 2026 DSC 240 Machine Learning

TA Winter 2025 DSC 148 Introduction to Data Mining

TA Spring 2023 DSC 258R Natural Language Processing

REVIEWER

ACL ARR, ICLR, CVPR, ICML, NeurIPS